

Lecture 6: The Normal Model

Professor Alexander Franks

2019-10-29

Announcements

- Reading: Chapter 5, Hoff

$$y \sim N(\mu, \sigma^2)$$

- 2 parameter model.

+ μ (mean)

+ σ^2 (variance)

Need $P(\mu, \sigma^2)$ (prior)

Get $P(\mu, \sigma^2 | y)$ (post.)

The Normal Distribution

- One of the most utilized probability models in data analysis
- Central Limit Theorem
- Separate parameters for the mean and the variance (intuitive)

$$Z_1, \dots, Z_n \sim \text{i.i.d.}$$

$$\bar{Z} \approx \text{normal}$$

Normal Distribution

- Symmetric with mode = median = mean = μ
- Approximately 95% of the population lies within two standard deviations of the mean $\pm 2\sigma$
- Density: Get $\mathcal{N}(\mu, \sigma)$

$$p(y|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}, \quad -\infty < y < \infty$$

- $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$ with X and Y independent then

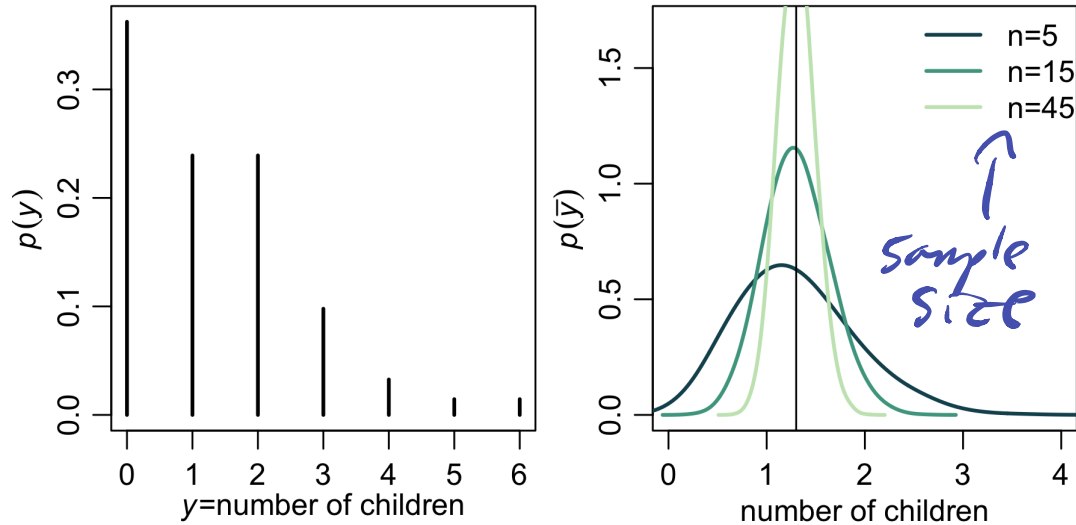
$$aX + bY \sim N(a\mu_x + b\mu_y, a^2\sigma_x^2 + b^2\sigma_y^2)$$

Get squared b/c var.

- In R: `dnorm`, `rnorm`, `pnorm`, `qnorm`.

- Warning: the argument to the `norm` functions in R is σ not σ^2 !

The Central Limit Theorem



$$Y \sim \text{Pois}(\lambda)$$

$$\text{CLT: } \bar{y} \approx N(E[Y], \text{Var}[Y]/n)$$

$$\bar{y} \approx N(1, \frac{1}{n})$$

Bayesian inference in the normal model

one-parameter

- Assume $y_1, \dots, y_n \sim N(\mu, \sigma^2)$ with σ^2 a known constant
- Lets start with a non-informative, improper prior: $p(\mu) \propto \text{const}$
- What is the posterior distribution $p(\mu \mid y_1, \dots, y_n, \sigma^2)$?

$\int_{-\infty}^{\infty} \text{const} = \infty$ but ... we will get a proper post.

$$P(\mu \mid \sigma^2, y_1, \dots, y_n) \propto L(\mu) \times \text{const.}$$

μ is unknown R.V.

σ^2 known const. (by assumption)

$$P(Y|u, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(Y-u)^2}{2\sigma^2}}$$

$$L(u) \propto e^{-\frac{(Y-u)^2}{2\sigma^2}}$$

Bayes:

Prior: $P(u) \propto \text{const.}$

$$P(u|y) \propto e^{f(u) + g(y)} \quad \text{not } u$$

$$\propto e^{f(u)} \cancel{e^{g(y)}}$$

$$\begin{aligned}
 P(\mu|\dots) &= L(\mu) \propto \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(y_i - \mu)^2}{2\sigma^2}} \\
 &= \left(\frac{1}{\sqrt{2\pi}\sigma^2} \right)^n e^{-\sum_{i=1}^n \frac{(y_i - \mu)^2}{2\sigma^2}} \quad (\text{expand the sq}) \\
 &\propto e^{-\frac{(\sum y_i^2 - 2\mu \sum y_i + n\mu^2)}{2\sigma^2}} \\
 &\propto \cancel{e^{-\frac{\sum y_i^2}{2\sigma^2}}} e^{\frac{2\mu \sum y_i - n\mu^2}{2\sigma^2}}
 \end{aligned}$$

Aside: Complete the sq.

$$\begin{aligned}
 (ax^2 - bx) &= a(x^2 - \frac{b}{a}x) \\
 &= a\left(x - \frac{b}{2a}\right)^2 - \frac{b^2}{4a}
 \end{aligned}$$

$$\begin{aligned}
 &e^{-\frac{(n\mu^2 - 2\mu \sum y_i)}{2\sigma^2}} \\
 &\propto e^{\frac{n(\mu - \frac{\sum y_i}{n})^2}{2\sigma^2}} \cancel{e^{-\frac{b^2}{4a}}}_{(no \mu)} \\
 &\propto e^{\frac{(\mu - \frac{\sum y_i}{n})^2}{2\sigma^2/n}} \quad P(\mu|\dots) \sim N\left(\bar{y}, \frac{\sigma^2}{n}\right)
 \end{aligned}$$

$$\begin{aligned}
 a &= n \\
 b &= 2\sum y_i \quad (\text{Dict}) \\
 x &= \mu
 \end{aligned}$$

$$P(\mu | Y, \sigma^2) \propto e^{-\frac{(\mu - \frac{\sum Y_i}{n})^2}{2\sigma^2/n}}$$

$$\propto e^{-\frac{(\mu - \bar{Y})^2}{2\sigma^2/n}}$$

$$P(\mu | Y_1, \dots, Y_n, \sigma^2) \sim N(\bar{Y}, \frac{\sigma^2}{n})$$

$$\sim E[\mu | Y, \sigma^2] = \bar{Y} = \hat{\mu}_{MLE}$$

$$Var(\mu | Y, \sigma^2) = \frac{\sigma^2}{n} = Var(\hat{\mu}_{MLE})$$

Matches Frequentist View

Bayesian inference in the normal model

- Assume $y_1, \dots, y_n \sim N(\mu, \sigma^2)$ with σ^2 a known constant
- The normal prior distribution is conjugate for μ in the normal sampling model
- Sampling distribution, prior distribution and posterior distribution are all normal.
- Assume the prior is $p(\mu) \sim N(\mu_0, \tau^2)$
- What are the parameters of the posterior $p(\mu \mid y_1, \dots, y_n, \sigma^2)$?

$$\begin{aligned}
 P(\mu | y_1, \dots, y_n, \sigma^2) &\propto L(\mu) \times P(\mu) \\
 &\propto \underbrace{e^{-\frac{(\bar{y} - \mu)^2}{2\sigma^2/n}}}_{L(\mu)} \underbrace{e^{-\frac{(\mu - \mu_0)^2}{2\tau^2}}}_{P(\mu)} \\
 &\propto \exp\left[-\frac{\bar{y}^2 - 2\bar{y}\mu + \mu^2}{2\sigma^2/n} + \frac{\mu^2 - 2\mu\mu_0 + \mu_0^2}{2\tau^2}\right] \\
 &\propto \exp\left[-\frac{1}{2} \left[\frac{n}{\sigma^2} \mu^2 - \frac{2n}{\sigma^2} \bar{y}\mu + \frac{1}{\tau^2} \mu^2 - \frac{2\mu\mu_0}{\tau^2} \right]\right] \\
 &\propto \exp\left[-\frac{1}{2} \left(\underbrace{\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)}_a \mu^2 - 2 \underbrace{\left(\frac{n\bar{y}}{\sigma^2} + \frac{\mu_0}{\tau^2}\right)}_b \mu \right)\right] \\
 &\propto \exp\left[-\frac{1}{2} \underbrace{\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)}_{1/\text{Var}} \left(\mu - \underbrace{\frac{\left(\frac{n\bar{y}}{\sigma^2} + \frac{\mu_0}{\tau^2}\right)}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}}_{\text{Mean}} \right)^2\right]
 \end{aligned}$$

$$P(\mu | y, \sigma^2) \propto N(\mu_n, \sigma_n^2)$$

$$\mu_n = \frac{\underbrace{\frac{n}{\sigma^2}}_w \underbrace{\bar{y}}_{\text{MLE}} + \underbrace{\frac{1}{\tau^2}}_{\text{Prior mean}} \underbrace{\mu_0}_{\text{Prior mean}}}{\underbrace{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}_{1-w}}$$

Posterior
Mean
of μ .

$$\sigma_n^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

$$\mu_n = w \bar{y} + (1-w) \mu_0$$

$$w = \frac{\frac{n}{\sigma^2}}{\frac{n}{\sigma^2} + \frac{1}{\tau^2}}$$

$\frac{n}{\sigma^2}$ is precision
of $\bar{y}$

$\frac{1}{\tau^2}$ is prior precision

A conjugate prior for the normal likelihood

- The normal distribution is conjugate for the normal likelihood
 - Often called the "normal-normal model"
- $Y_i \sim N(\mu, \sigma^2)$ and $\mu \sim N(\mu_0, \tau^2)$ implies that the posterior distribution $p(\mu \mid y)$ is also normally distributed:

$$\mu \mid Y \sim N(\mu_n, \tau_n^2)$$

$$\text{where } \mu_n = \frac{\frac{1}{\tau^2} \mu_0 + \frac{n}{\sigma^2} \bar{y}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}} \text{ and } \tau_n^2 = \frac{1}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$$

The posterior mean and pseudo-counts

$$\begin{aligned}\mu_n &= \frac{\tau^2}{\tau^2 + n\sigma^2} \mu_0 + \frac{n\sigma^2}{\tau^2 + n\sigma^2} \bar{y} \\ &= (1 - w) \mu_0 + w \bar{y}\end{aligned}$$

where $w = \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$

$$w = \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$$

Can we think about the normal prior parameters in terms of pseudo-counts?

$$\tau^2 = \frac{\sigma^2}{K_0} \quad w = \frac{n}{n + K_0}$$

K_0 is # of "prior observations"

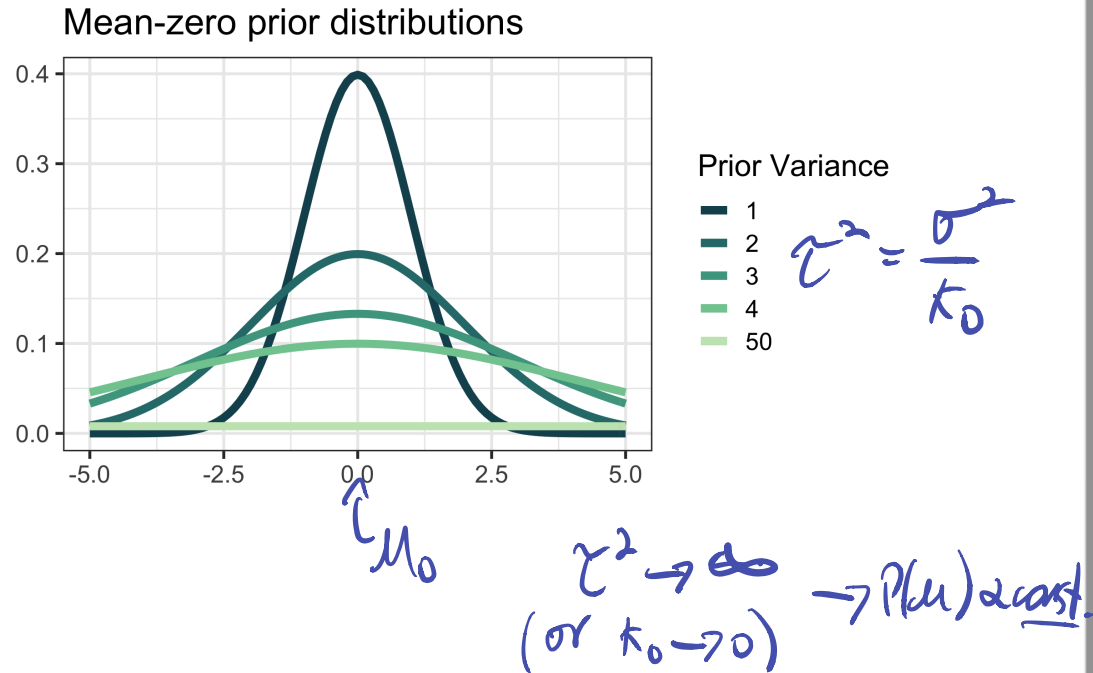
The posterior mean and pseudo-counts

$$\begin{aligned}\mu_n &= \frac{\tau^2}{\tau^2 + n\sigma^2} \mu_0 + \frac{n\sigma^2}{\tau^2 + n\sigma^2} \bar{y} \\ &= (1 - w) \mu_0 + w \bar{y}\end{aligned}$$

where $w = \frac{\frac{n}{\sigma^2}}{\frac{1}{\tau^2} + \frac{n}{\sigma^2}}$

- Let's reparameterize: $\tau^2 = \frac{\sigma^2}{\kappa_0}$
- Then: the posterior variance is $\frac{\sigma^2}{\kappa_0 + n}$
- And: $(1 - w) = \frac{\kappa_0}{\kappa_0 + n}$
- κ_0 are the prior counts and μ_0 is the prior sample average.

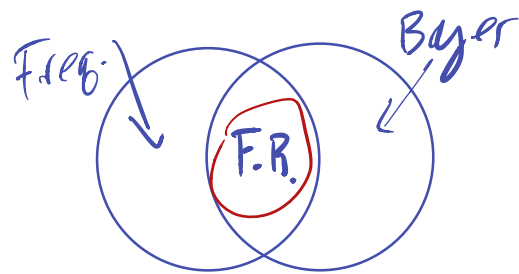
Conjugate prior with increasing variance



Bayes Estimators

$$E[\mu | Y_1, \dots, Y_n, \sigma^2] = w\bar{Y} + (1-w)\mu_0$$

What are the frequentist properties of $E[\mu | Y_1, \dots, Y_n, \sigma^2]$



Estimate.
(const).

Think about $\underbrace{w\bar{Y} + (1-w)\mu_0}_{\text{Estimator}}$

Estimators: Bayes / Frequentist Unification

- Bayesian inference provides a straightforward procedure for producing estimators given your prior beliefs.
 1. Compute posterior distribution
 2. Summarize the posterior distribution with a point estimator (e.g. posterior mean or posterior mode) and a probability interval
- Frequentists provide tools for evaluating the sampling properties of an estimator.
 - Bias, variance and MSE of an estimator
 - Well-calibrated probability intervals
- Both are useful!

The Bias-Variance Tradeoff

Reminder: an estimator is a random variable, an estimate is a constant

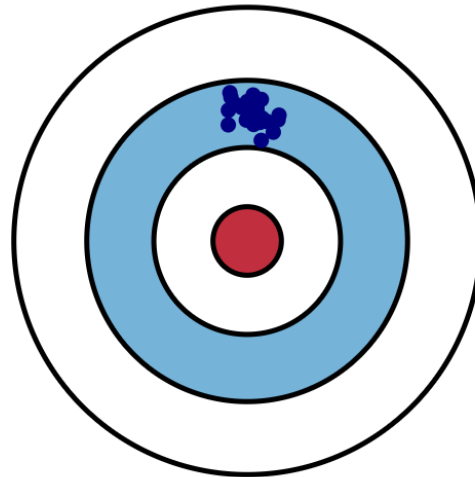
- *Bias*: systematic sampling error of the estimator
- *Variance*: variance of the estimator (from sampling & measurement error)
- Often we evaluate an estimator in terms of mean square error:
$$\text{MSE}(\hat{\theta}) = E_Y(\hat{\theta} - \theta)^2$$
- The Bias-Variance tradeoff:
$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

The Bias-Variance Tradeoff

- Variance of an estimator comes sampling from a population
 - If you were to repeatedly draw new samples of the same size how much would your estimates vary?
 - e.g. if $y_i \sim N(\mu, \sigma^2)$ then $\text{Var}(\bar{Y}) = \sigma^2/n$

Bias

The expected difference between the estimate and the response

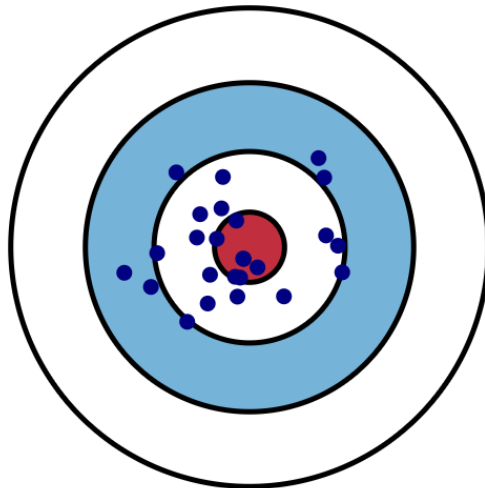


Statistical definition of bias:

$$E_Y[\hat{\theta} - \theta]$$

Variance

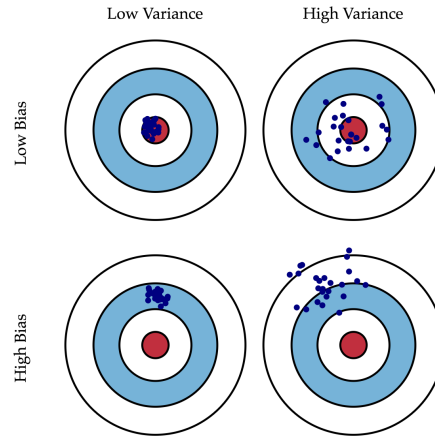
How variable is the prediction about its mean?



Statistical definition of variance:

$$E_Y[\hat{\theta} - E_Y[\hat{\theta}]]^2$$

Bias and Variance



$$\underbrace{\text{MSE}(\hat{\theta})}_{\text{accuracy}} = \underbrace{\text{Var}(\hat{\theta})}_{\text{variance}} + \underbrace{\left(E[\hat{\theta} - \theta] \right)^2}_{\text{bias squared}}$$

The Bias-Variance Tradeoff

- The prior distribution (usually) makes your estimator biased...
- But the prior distribution also (usually) reduces the variance!
- Example: compute the frequentist mean and variance of the posterior mean.

$$\text{Bias}_{MLE} E[\bar{Y} - \mu] = E[\bar{Y}] - \mu = 0$$

$$\text{Var}_{MLE}(\bar{Y}) = \text{Var}\left(\frac{\sum Y_i}{n}\right) = \frac{1}{n^2} \text{Var}(\sum Y_i) = \frac{\sigma^2}{n}$$

$$MSE: \text{Bias}^2 + \text{Var} = \frac{\sigma^2}{n}$$

0 for
MLE

$$\hat{\mu}_{PM} = w\bar{Y} + (1-w)\mu_0$$

Bias:

$$E[w\bar{Y} + (1-w)\mu_0 - \mu] =$$

$$wE[\bar{Y}] + (1-w)\mu_0 - \mu =$$

$$w\mu + (1-w)\mu_0 - \mu =$$

$$\underline{(1-w)(\mu_0 - \mu)} \quad \text{Bias}$$

$$\text{Var}(w\bar{Y} + (1-w)\mu_0) =$$

$$w^2 \text{Var}(\bar{Y}) + (1-w)^2 \cancel{\text{Var}(\mu_0)}$$

$$\underline{w^2 \text{Var}(\bar{Y})} < \text{Var}(\bar{Y})$$

Var